

Question ID: 369b7bb7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function g is defined by $g(x) = \sqrt{8x + 1}$. What is the value of $g(3)$?

Answer

- A. $\frac{5}{8}$
- B. $\frac{25}{8}$
- C. 5
- D. 25

Correct Answer: C

Rationale

Choice C is correct. It’s given that the function g is defined by $g(x) = \sqrt{8x + 1}$. Substituting 3 for x in the given function yields $g(3) = \sqrt{8(3) + 1}$, which is equivalent to $g(3) = \sqrt{25}$, or $g(3) = 5$. Therefore, the value of $g(3)$ is 5 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $8(3) + 1$, not $\sqrt{8(3) + 1}$.